

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 3: TISSUES IN ACTION [SET C - NCERT EXEMPLAR SPECIAL]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH03/SET-C****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET C - NCERT EXEMPLAR & CRITICAL THINKING):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** NCERT Exemplar based Multiple Choice Questions with single correct option.
3. **Section B (Q16 – Q20):** Deep Assertion-Reasoning based questions testing scientific cause-and-effect.
4. **Section C (Q21 – Q25):** Case-Study / Laboratory Experimental Investigation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: NCERT EXEMPLAR MCQS (Q.1 TO Q.15)**[15 MARKS]****Q.1 During daytime, what physiological event triggers the opening of stomatal pores in green plant leaves?**

- (A) Active influx of potassium (K^+) ions into guard cells, followed by endosmotic water intake that makes guard cells turgid
- (B) Loss of water from guard cells causing flaccidity
- (C) Synthesis of thick lignin in guard cell walls
- (D) Accumulation of starch in guard cell vacuoles

Q.2 A student places the freshly cut stem of a white carnation flower in a beaker of red eosin dye solution. Within hours, red streaks appear along the flower petals. Which tissue is stained red?

- (A) Phloem sieve tubes
- (B) Xylem vessels and tracheids
- (C) Cortex parenchyma
- (D) Pith collenchyma

Q.3 Which component of human blood lacks a nucleus at maturity, possesses a biconcave disc shape, and has an average lifespan of about 120 days?

- (A) Monocyte
- (B) Erythrocyte (Red Blood Cell)
- (C) Neutrophil
- (D) Thrombocyte (Platelet)

Q.4 At a chemical synapse, the arrival of an electrical action potential at the axon terminal triggers the release of which chemical into the synaptic cleft?

- (A) Hemoglobin
- (B) Neurotransmitter (such as acetylcholine)
- (C) Insulin
- (D) Collagen fibres

Q.5 Which tissue lines the inner surfaces of the human fallopian tubes (oviducts) to propel the non-motile ovum towards the uterus?

- (A) Ciliated columnar epithelium
- (B) Stratified squamous epithelium
- (C) Simple cuboidal epithelium
- (D) Adipose connective tissue

Q.6 The microscopic structural and functional unit of mammalian skeletal muscle consisting of repeating actin and myosin myofilament bands between two Z-discs is termed:

- | | |
|----------------|---------------|
| (A) Sarcolemma | (B) Sarcomere |
| (C) Sarcoplasm | (D) Fascicle |

Q.7 Which type of connective tissue forms the internal framework (stroma) that supports lymphoid organs such as spleen, lymph nodes, and bone marrow?

- | | |
|---------------------------------|--------------------------------------|
| (A) Reticular connective tissue | (B) Dense regular collagenous tissue |
| (C) Hyaline cartilage | (D) Elastic cartilage |

Q.8 In woody perennial trees, the living cells of the bark obtain atmospheric oxygen for respiration through porous aerating openings called:

- | | |
|----------------|----------------------|
| (A) Stomata | (B) Lenticels |
| (C) Hydathodes | (D) Casparian strips |

Q.9 Which animal tissue cells are capable of continuous mitotic division throughout an adult's life to regenerate lost epidermis and intestinal lining?

- | | |
|---------------------------------|--------------------------------|
| (A) Cardiac myocytes | (B) Neurons in cerebral cortex |
| (C) Epithelial basal stem cells | (D) Mature osteocytes |

Q.10 Which component of phloem is primarily responsible for the lateral conduction of food and water in stems?

- | | |
|---------------------|-------------------------------------|
| (A) Sieve tubes | (B) Phloem parenchyma / Phloem rays |
| (C) Companion cells | (D) Bast fibres |

Q.11 The white glistening fibers that endow tendons with immense tensile strength along the longitudinal axis of pull are composed of:

- | | |
|-------------|-----------------------|
| (A) Elastin | (B) Collagen (Type I) |
| (C) Keratin | (D) Actin |

Q.12 Why are cartilage rings shaped like a 'C' present along the human trachea?

- | | |
|--------------------------------------|---|
| (A) To filter inhaled dust particles | (B) To prevent the tracheal airway from collapsing during inhalation when thoracic pressure drops |
| (C) To produce vocal sounds | (D) To warm inhaled air |

Q.13 Which blood cell type plays the central role in cellular immunity and antibody-mediated defense against microbial pathogens?

- | | |
|---------------------------------------|---------------|
| (A) Erythrocytes | (B) Platelets |
| (C) Lymphocytes (T-cells and B-cells) | (D) Basophils |

Q.14 Which plant tissue exhibits thick-walled, dead, elongated cells with tapering ends and narrow lumen, commonly extracted commercially from jute and hemp?

- | | |
|-----------------|---------------------------------------|
| (A) Collenchyma | (B) Sclerenchyma fibres (bast fibres) |
| (C) Parenchyma | (D) Xylem parenchyma |

Q.15 The myelin sheath surrounding peripheral nerve axons is produced by which specialized glial cells?

- | | |
|----------------|---------------------------|
| (A) Astrocytes | (B) Schwann cells |
| (C) Microglia | (D) Oligodendrocytes only |

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): Transpiration pull is the major driving force for the ascent of water in tall coniferous trees exceeding 100 meters.

Reason (R): Cohesion between water molecules and adhesion to xylem walls maintain an unbroken capillary water column under tension.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): Involuntary smooth muscles are found in the walls of blood vessels and bronchi.

Reason (R): Smooth muscle contractions help regulate blood pressure and airflow without conscious voluntary intervention.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): Intercellular matrix is absent in epithelial tissue.

Reason (R): Epithelial cells are compactly packed on a non-cellular basement membrane with specialized cell junctions.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Companion cells are present in gymnosperm phloem.

Reason (R): Gymnosperms possess sieve cells and albuminous cells instead of sieve tubes and companion cells.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): Blood platelets (thrombocytes) are essential for blood coagulation.

Reason (R): Platelets aggregate at injury sites and release thromboplastin to initiate the clotting cascade.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL INVESTIGATION & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL INVESTIGATION:

Case Study: Microscopic Examination of Plant Meristematic Zones

A biology student prepares a longitudinal section (LS) of an onion root tip stained with acetocarmine. Under the microscope, she identifies four distinct longitudinal zones from the tip upward:

- Zone I (Apex):** A thimble-like cap of loose parenchymatous cells covering the extreme apex.
- Zone II (Meristematic zone):** Densely packed small isodiametric cells with thin cellulose walls, dense granular cytoplasm, and large nuclei, with several cells showing mitotic chromosomes.
- Zone III (Elongation zone):** Cells rapidly increasing in length with large central vacuoles forming.
- Zone IV (Maturation zone):** Cells developing thickened walls and forming root hairs.

Q.21 What is the primary function of Zone I (Root cap)?

- (A) Absorbing mineral fertilizers
(B) Protecting the tender apical meristem from mechanical abrasion as the root pushes through rough soil
(C) Conducting manufactured sugars
(D) Performing photosynthesis

Q.22 Why were mitotic dividing figures observed exclusively in Zone II?

- (A) Zone II contains the root apical meristem where active continuous mitotic cell division takes place
(B) Cells in Zone II are dead
(C) Zone II is where flowers grow
(D) Mitosis only occurs in root caps

Q.23 Which zone is primarily responsible for the rapid linear growth and extension of the root into deep soil?

- (A) Zone I (Root cap)
(B) Zone II (Meristematic zone)
(C) Zone III (Zone of elongation)
(D) Zone IV (Root hair zone)

Q.24 Root hairs observed in Zone IV are cellular extensions of which tissue layer?

- (A) Cortex
(B) Endodermis
(C) Epiblema (Rhizodermis / Root epidermis)
(D) Pericycle

Q.25 In which zone do cells differentiate into specialized vascular tissues (xylem vessels and phloem sieve tubes)?

- (A) Zone I
(B) Zone II
(C) Zone III
(D) Zone IV (Zone of maturation / differentiation)

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & EXEMPLAR EXPLANATIONS (SET C)

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.1	(A)	$\Delta K^+ + \Delta$ influx lowers the water potential inside guard cells; water enters by endosmosis. Turgid guard cells bend outward along their thin outer walls, opening the pore.
Q.2	(B)	Xylem is the conducting tissue for water and dissolved minerals; transpirational pull draws the red eosin dye upward through xylem conduits.
Q.3	(B)	Mature human erythrocytes extrude their nuclei to maximize space for hemoglobin, maintaining a biconcave disc shape for efficient gas exchange over their 120-day life.
Q.4	(B)	Action potentials cause synaptic vesicles to fuse with the presynaptic membrane, releasing neurotransmitters (like acetylcholine) to diffuse across the synaptic cleft.
Q.5	(A)	Coordinated rhythmic beating of cilia on the luminal surface of ciliated columnar epithelial cells drives the ovum forward along the oviduct.
Q.6	(B)	A sarcomere is the basic contractile unit of striated muscle bounded between adjacent Z-lines.
Q.7	(A)	Reticular tissue consists of delicate networks of type III collagen reticular fibers and fibroblasts, supporting free immune cells in lymphoid organs.
Q.8	(B)	Lenticels are lens-shaped porous regions in the periderm of woody stems that facilitate gaseous exchange across suberized cork layers.
Q.9	(C)	Basal epithelial stem cells divide continuously by mitosis to replace sloughed-off superficial cells of skin and mucous membranes.
Q.10	(B)	Phloem ray parenchyma cells conduct nutrients and water radially (laterally) across the stem cross-section.
Q.11	(B)	Tendons are composed of parallel bundles of Type I collagen fibers, providing extraordinary tensile strength along the direction of muscular traction.
Q.12	(B)	Incomplete C-shaped rings of hyaline cartilage keep the trachea permanently patent (open) and prevent collapse during negative pleural pressure breathing.
Q.13	(C)	Lymphocytes (B cells producing antibodies, T cells performing cell-mediated lysis) are the key orchestrators of adaptive immune defense.
Q.14	(B)	Commercial jute, hemp, and flax fibers are bast sclerenchyma fibers harvested from phloem and pericycle for textile and rope manufacture.
Q.15	(B)	In the peripheral nervous system (PNS), Schwann cells wrap repeatedly around axons to form insulating myelin sheaths.
Q.16	(A)	The cohesion-tension theory explains that transpirational evaporation pulls unbroken columns of water upward due to strong hydrogen bonding (cohesion) between water molecules.
Q.17	(A)	Vascular and bronchial smooth muscle responds automatically to autonomic and hormonal signals (vasoconstriction/vasodilation) to sustain homeostatic perfusion.
Q.18	(A)	Epithelial sheets form continuous protective barriers where adjacent cells are tightly bound by desmosomes and tight junctions with negligible extracellular matrix.
Q.19	(D)	Assertion is false: Gymnosperms lack true companion cells and sieve tubes. Reason is true: Gymnosperms have sieve cells associated with Strasburger (albuminous) cells.
Q.20	(A)	Platelets form a mechanical hemostatic plug and release clotting factors that convert prothrombin into thrombin, generating an insoluble fibrin clot.
Q.21	(B)	The root cap protects the fragile actively dividing apical meristem behind it and secretes mucilage to lubricate root penetration into soil.

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.22	(A)	The apical meristem in Zone II consists of undifferentiated, perpetually dividing cells responsible for primary root growth.
Q.23	(C)	Cellular elongation and vacuolation in Zone III exert mechanical pushing force that drives root lengthening deep into the earth.
Q.24	(C)	Root hairs are unicellular tubular outgrowths of epidermal cells (epiblema/rhizodermis) that vastly amplify surface area for water and mineral absorption.
Q.25	(D)	In the maturation zone, elongated cells undergo cytological differentiation into mature permanent tissues: xylem, phloem, cortex, and endodermis.